Customer Shopping Behavior Analysis

Insights from 3,900 purchases across product categories

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Dataset Overview

3.9K

Transactions

Purchase records analyzed

18

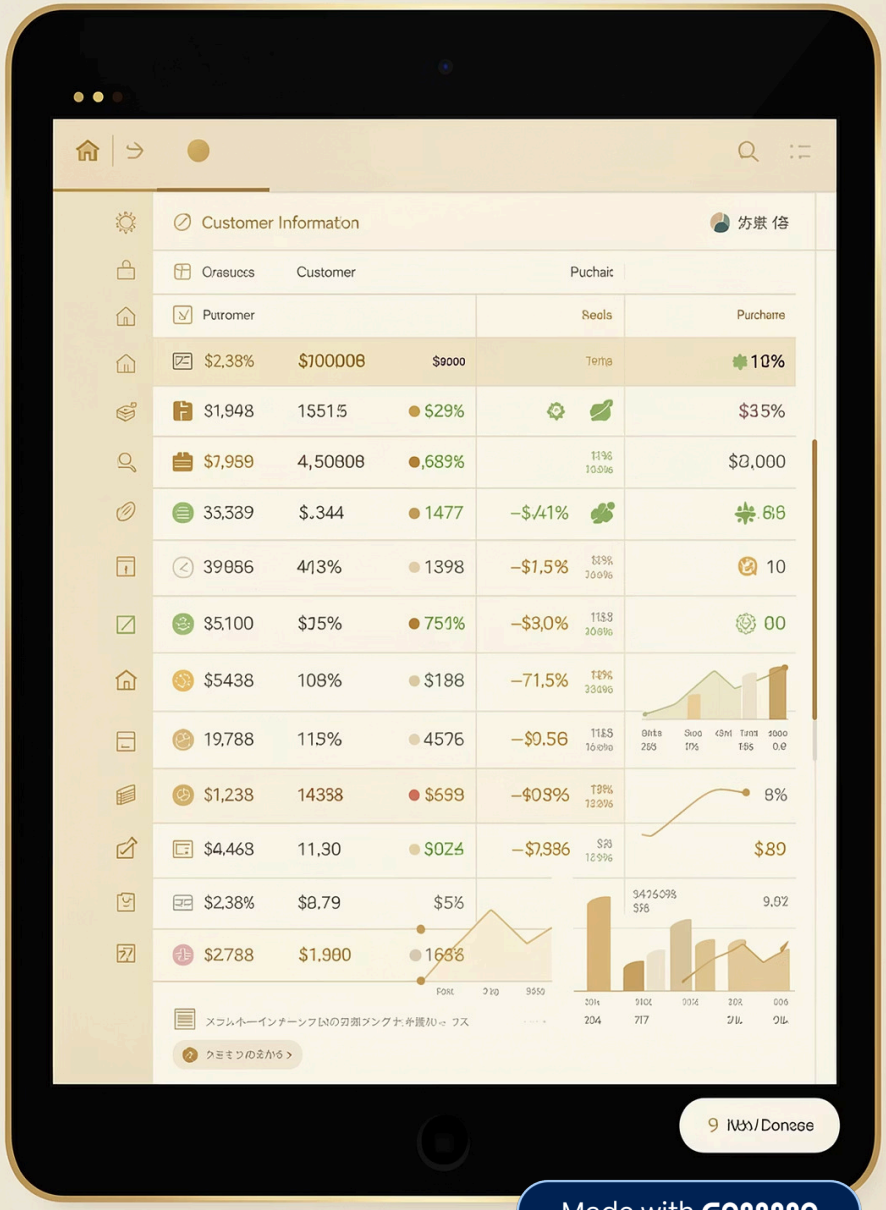
Features

Data points per transaction

37

Missing Values

Review ratings imputed



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Python Data Preparation

Data Loading & Exploration

- Imported with pandas
- Checked structure with info()
- Summary statistics generated

Data Cleaning Steps

- Imputed missing ratings
- Standardized column names
- Dropped redundant fields

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Feature Engineering

Age Groups

Binned customer ages into segments

Purchase Frequency

Calculated days between purchases

Database Load

Connected to PostgreSQL

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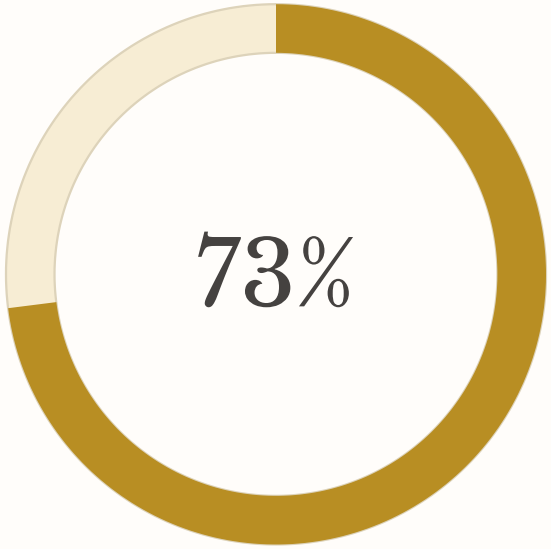
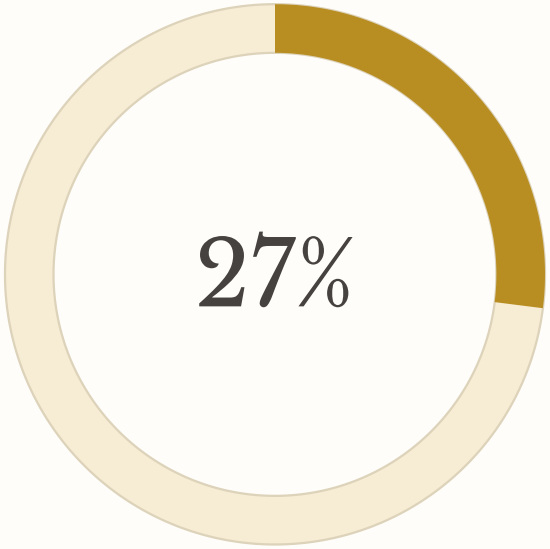
SQL Analysis: Revenue Insights



Male Revenue

$157,890 total



Female Revenue

$75,191 total

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Product Performance

Top-Rated Products

1. Gloves (3.86)
2. Sandals (3.84)
3. Boots (3.82)
4. Hat (3.80)
5. Skirt (3.78)

Most Ordered

- Blouse: 171 orders
- Pants: 171 orders
- Jewelry: 171 orders
- Coat: 161 orders

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Discount Strategy Analysis

High-Spending Discount Users

839 customers used discounts but spent above average

Most Discounted Products

Hat (50%), Sneakers (49.66%), Coat (49.07%)

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Customer Segmentation

1

Loyal

3,116 customers

2

Returning

701 customers

3

New

83 customers

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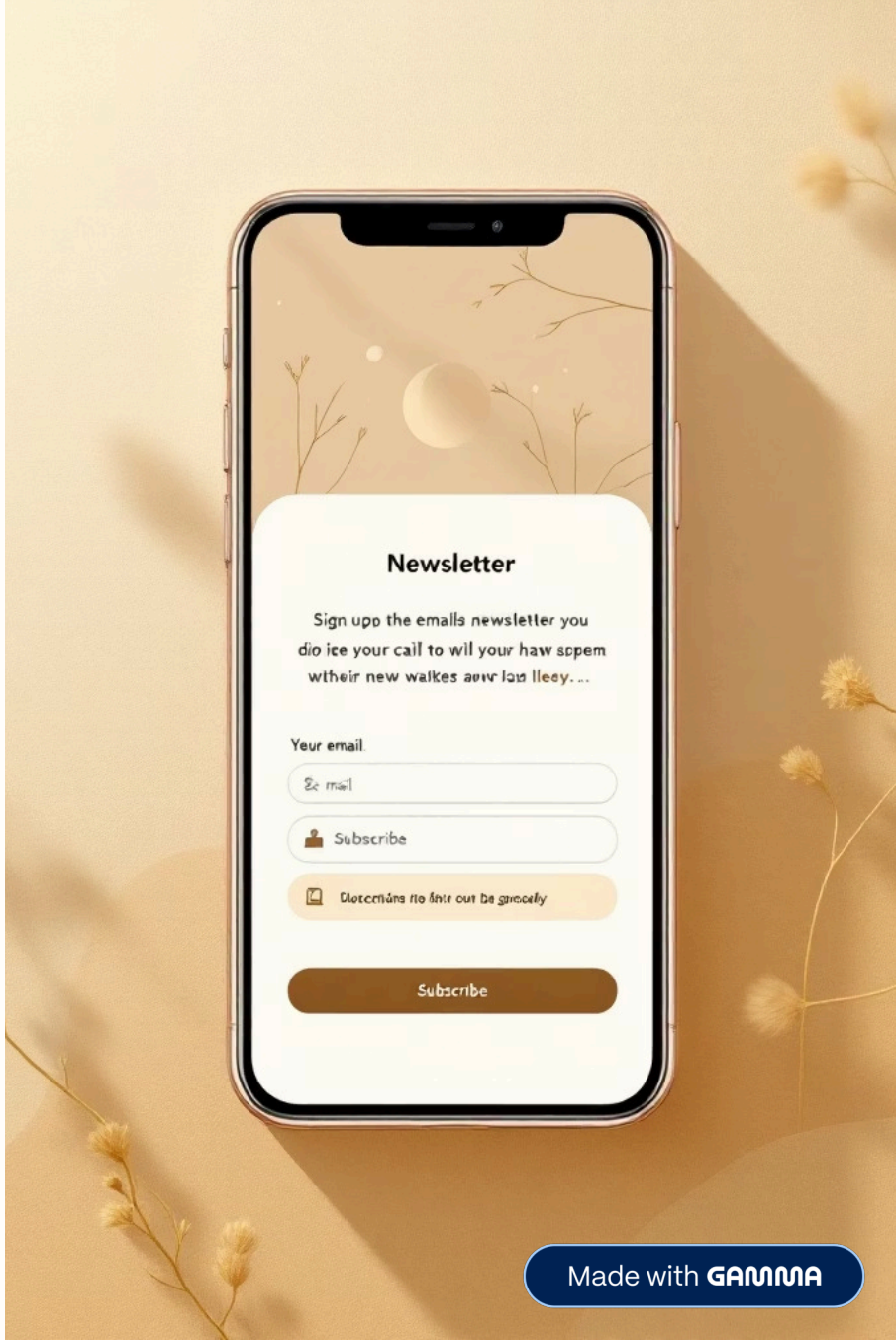
Subscription Impact

Subscriber Stats

- 1,053 subscribers
- Avg: $59.49
- Total: $62,645

Non-Subscriber Stats

- 2,847 non-subscribers
- Avg: $59.87
- Total: $170,436



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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits

Customer Loyalty

Reward repeat buyers

Targeted Marketing

Focus on high-revenue groups

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